

Throttling Protocol for Quantum Links

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Abstract We propose a new throttling mechanism for quantum link control in entanglement-based quantum networks. Using Markov chain modeling, we demonstrate Bell pair delivery stabilization, enabling an easier consumption by the upper layers of the quantum network protocol stack. ©2026 The Author(s)

Introduction

Quantum communications began with the invention of quantum key distribution (QKD) in 1984 by Bennet and Brassard [1]. Because of the no-cloning theorem [2], one cannot amplify the quantum signal. Thus, current QKD systems are typically limited to 100 km of single-mode fiber because of fiber losses. To extend them, a first solution is to use intermediate trusted nodes where the quantum nature of communications is broken, but these systems are inherently less secure, as the key is available in each trusted node. A better solution to enforce end-to-end secure communications based on the laws of quantum physics is to maintain the quantum nature of the communications using quantum repeaters. This will not only secure QKD over longer distances, it will also enable other quantum applications that are not feasible with classical networks [3]: distributed quantum computing, blind quantum computing, quantum sensing and metrology applications, fast Byzantine negotiation, etc.

By applying quantum teleportation [4] to Bell pairs (BPs), quantum nodes can propagate these BPs over the network. We call this operation *entanglement stitching* (a.k.a. *entanglement swapping*). The role of quantum networks can be reduced to distributing BPs between any pair of nodes [4]. Indeed, quantum applications running on top of these entanglement-based quantum networks will consume BPs to transmit quantum states using quantum teleportation or to apply distributed quantum operations.

Because the hardware is not yet mature, the road towards entanglement-based networks remains long, despite some experimental proofs of concepts [5–8]. Their control and management brings new interesting challenges for optical and networking researchers [9–13]. Solutions for the quantum link protocol (QLP) have been proposed [11–13], but none aim to mitigate the jitter of BP delivery due to fiber losses.

In this paper, after describing how quantum links could be realized, we propose a throttling mechanism to remove the jitter of BP delivery by the QLP, we model its behavior using Markov chains, and we evaluate its performances using this model.

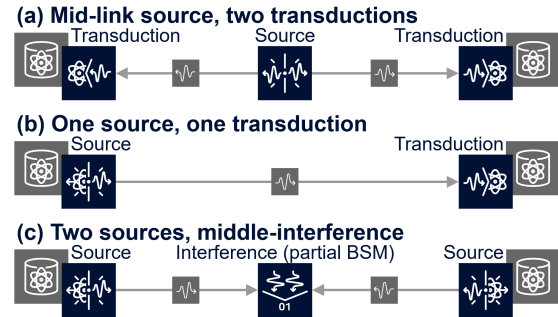


Fig. 1: Various realizations of a quantum link.

Physical link realization

The aim of a quantum link is to create BPs between neighboring nodes. This requires the creation of BPs, the link transmission of at least one qubit (photon) of each BP and the storage of qubits in quantum memories (matter qubits) to share entanglement between these nodes. Fig. 1 gives three different ways to realize a quantum link:

- A source in the middle of the link creates a BP of photons that are each sent to a different node, where they are transduced to matter qubits in quantum memories;
- A source in one node creates a BP comprising a matter qubit and a photon, the latter being sent to the other node where it is transduced to a matter qubit in its quantum memory;
- Two sources, one in each node, create BPs comprising a matter qubit and a photon, the latter being sent to an optical interferometer, which performs a partial Bell State Measurement (BSM), effectively transforming the matter qubits into a BP.

In all cases, the success or failure of BP creation, fiber transmission, qubit transduction or interference is heralded by acknowledgement messages sent to the nodes at the link extremities. We note η_{QLL} and $\eta_{QLS} = 1 - \eta_{QLL}$ the proportions of lost BPs and of successfully established BPs, i.e., BPs for which both qubits have been successfully received and stored in the quantum memories of their respective nodes.

Experimental implementations of quantum links have been realized [5–8]. The performances are not yet good enough for practical quantum networks, but we can expect future improvements.

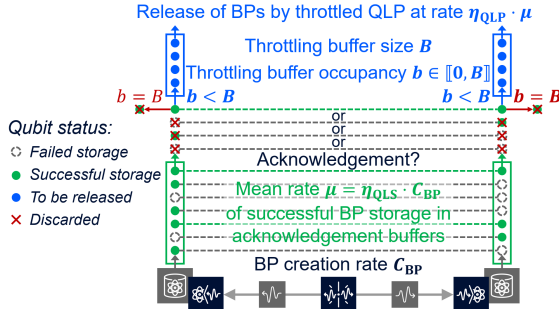


Fig. 2: Throttled quantum link protocol.

Throttled QLP proposal

Fig. 2 illustrates our throttled QLP proposal in the case of mid-link sources (Fig. 2.a). Other cases work similarly, though the parameters may vary. Quantum memories are required in each node to store received qubits while waiting for the acknowledgement that the other qubit in the BP was properly transmitted and stored: we call *acknowledgement buffer* the part of the quantum memory dedicated for this process. Its size depends on the BP creation rate C_{BP} and the time delay between the reception of the qubit in each node and the reception of the corresponding acknowledgement message, and may differ between both nodes.

If the acknowledgement signals a failure, the qubit in the buffer is discarded. Otherwise, it is further processed. But the processes that causes a failure, such as fiber losses, induce random losses on qubits. Thus, the amount of BPs to be further processed over a short period of time Δt varies from 0 to $C_{BP} \cdot \Delta t$, averaging around $\mu \cdot \Delta t$, where $\mu = \eta_{QLS} \cdot C_{BP}$ is their average rate. Given this high volatility, we need a mechanism to stabilize the release of BPs by the QLP to the upper layers of the quantum network protocol stack. What we propose is to throttle this release, i.e. limit its rate to $\eta_{QLP} \cdot \mu$ with throttling ratio $\eta_{QLP} < 1$, using an additional *throttling buffer* of size B to store processable qubits before releasing them. We will show latter that the size B of this throttling buffer can be small.

If a qubit has just been acknowledged while the throttling buffer is full, it is instead discarded. If the throttling buffer is empty when the QLP should release a bell pair, the delivery of the BP by the QLP fails. This occurs with probability ϵ .

The nodes can be synchronized such that the discards and failures of the throttling process coincide (up to a constant delay), in effect discarding qubits from the same BPs. Due to lack of space, we do not detail how to do it in this paper.

Markov chain modeling of throttled QLP

We model the behavior of our throttled QLP proposal with a Markov chain. For the theory of Markov chains, see for example Brémaud [14, §2-4]. We use the following mathematical notation: $\forall k \leq l \in \mathbb{N}, \llbracket k, l \rrbracket = \{m \in \mathbb{N} \mid k \leq m \leq l\}$,

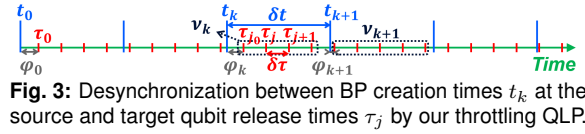


Fig. 3: Desynchronization between BP creation times t_k at the source and target qubit release times τ_j by our throttling QLP.

BPs are generated every $\delta\tau$ time unit. Thus, the storage process in the acknowledgement buffers is a Bernoulli arrival process with interval time $\delta\tau$ and probability of arrival η_{QLS} . The BP successful storage rate is $\mu = \frac{\eta_{QLS}}{\delta\tau}$. The possible arrival times are $\tau_j = \tau_0 + j \cdot \delta\tau$ with $j \in \mathbb{N}$.

The throttled QLP tries to release qubits to the upper layers every $\delta t = \frac{\delta\tau}{\eta_{QLP} \cdot \eta_{QLS}}$ time unit. If there are no qubits in the throttling buffer, the release fails. This happens with probability ϵ , which should be close to 0. The possible release times are $t_k = t_0 + k \cdot \delta t$ with $k \in \mathbb{N}$.

For time t_k , the throttling buffer occupancy just before the potential release of a qubit to the upper layers is b_k . If $b_k > 0$, the QLP can release a qubit, whereas if $b_k = 0$, it cannot. This occurs with probability $\epsilon_k = \text{Prob}[b_k = 0]$.

Fig. 3 shows how the two processes may be desynchronized. The first possible arrival after t_k occurs at time $\tau_{j_0(k)}$, with a phase offset of $\varphi_k = \tau_{j_0(k)} - t_k \in [0, \delta\tau[$. The last possible arrival before t_{k+1} occurs at time $\tau_{j_0(k)+\nu_k-1}$, where $\nu_k = \frac{\delta t + \varphi_{k+1} - \varphi_k}{\delta\tau}$ is the number of possible storages during the time interval $[t_k, t_{k+1}]$. The values ν_k and φ_k can be recursively calculated with:

$$\nu_k = \left\lceil \frac{\delta t - \varphi_k}{\delta\tau} \right\rceil \in \left\{ \nu_- = \left\lfloor \frac{\delta t}{\delta\tau} \right\rfloor, \nu_+ = \left\lceil \frac{\delta t}{\delta\tau} \right\rceil \right\} \quad (1)$$

and:

$$\varphi_{k+1} = \varphi_k + \delta\tau \cdot \nu_k - \delta t. \quad (2)$$

Denote a_k the number of successful storage operations during $[t_k, t_{k+1}]$, which follows the binomial probability distribution $bp_k = \text{Bin}(\nu_k, \eta_{QLS})$, i.e. such as, $\forall a \in \llbracket 0, \nu_k \rrbracket$:

$$bp_k(a) = \binom{\nu_k}{a} \cdot \eta_{QLS}^a \cdot (1 - \eta_{QLS})^{\nu_k - a}, \quad (3)$$

$bp_k(a) = 0$ otherwise. Note that bp_k is one of two distributions, bp_- or bp_+ , depending on whether ν_k is ν_- or ν_+ .

Let π_k be the probability distribution vector of b_k . We have $\pi_{k+1}^T = \pi_k^T \times M^{(k)}$, where $M^{(k)}$ is one of two transition matrices M^- or M^+ , depending on whether ν_k is ν_- or ν_+ :

$$\begin{aligned} M^{(k)} = M^\pm &= \left(M_{b,b'}^{(\pm)} \right)_{(b,b') \in \llbracket 0, B \rrbracket^2} \text{ with:} \\ \circ \forall b' \in \llbracket 0, B-1 \rrbracket, M_{0,b'}^{(\pm)} &= pb_\pm(b'), \\ \circ M_{0,B}^{(k)} &= \sum_{\beta \geq B} pb_\pm(\beta), \\ \circ \forall b \in \llbracket 1, B \rrbracket, \forall b' \in \llbracket 0, B-1 \rrbracket, \\ M_{b,b'}^{(\pm)} &= pb_\pm(b' - b + 1), \\ \circ \forall b \in \llbracket 1, B \rrbracket, M_{b,B}^{(\pm)} &= \sum_{\beta \geq B-b+1} pb_\pm(\beta). \end{aligned} \quad (4)$$

Because $M^{(k)}$ depends on ν_k , the process b_k with $k \in \mathbb{N}$ is a non-homogeneous Markov chain [14, def. 2.1.1], except in the rare cases when $\nu_+ = \nu_-$, i.e. when $\frac{\delta t}{\delta \tau}$ is an integer. To extract a homogeneous Markov chain from this, we approximate $\frac{\delta t}{\delta \tau}$ by a rational number, i.e. we suppose that $\frac{\delta t}{\delta \tau} = \frac{n}{c} \in \mathbb{Q}$, with $n, c \in \mathbb{N} \setminus \{0\}$. With this approximation, $c \cdot \delta t = n \cdot \delta \tau$, which implies that, $\forall k \in \mathbb{N}$, we have exactly $n = \sum_{l=k}^{k+c-1} \nu_l$ possible arrivals and c possible departures in $[t_k, t_{k+c}]$.

Thus, $\forall \kappa \in [0, c-1]$, the process $b(t_{\kappa+l \cdot c})$ with $l \in \mathbb{N}$ can be modeled by a homogeneous Markov chain $MC^{(\kappa)}$ [14, def. 2.1.1], with the following transition probability matrix:

$$M^{[\kappa]} = \prod_{m=\kappa}^{\kappa+c-1} M^{(m)}, \quad (5)$$

where, the $M^{(m)}$ depends only on the ν_m , which can be recursively calculated with Eq. (1) and (2). The $MC^{(\kappa)}$ differ, as they depend on their respective phase offset φ_κ .

One can check that each Markov chain $MC^{(\kappa)}$ is irreducible [14, def. 2.3.5]. Because it is a finite Markov chain, it is also positive recurrent [14, th. 3.2.8]. Thus, it admits the stationary probability distribution $\tilde{\pi}_\kappa$ [14, th. 3.2.6] such that:

$$\tilde{\pi}_\kappa^T = \tilde{\pi}_\kappa^T \times M^{[\kappa]}. \quad (6)$$

Furthermore, we have that:

$$\tilde{\pi}_\kappa^T = \tilde{\pi}_0^T \times \prod_{l=0}^{\kappa-1} M^{(l)}, \quad (7)$$

Because a release failure occurs at time t_k where $b_k = 0$, the probability of a QLP release failure under the stationary distribution is:

$$\epsilon = \frac{\sum_{\kappa=0}^{c-1} \tilde{\pi}_\kappa(0)}{c}. \quad (8)$$

Throttled QLP performance analysis

We created a Python program that calculates the transition matrix $M^{[0]}$ using Eq. (1–5). To solve Eq. (6), it multiplies an initial distribution π_0 many times by $M^{[0]}$, allowing for rapid convergence to the stationary distribution $\tilde{\pi}_0$ [14, §4.3.2]. Finally, it calculates the failure rate ϵ using Eq. (7–8).

Fig. 4 shows ϵ for different throttling buffer sizes B and throttling ratios, taking the quantum link loss η_{QLL} to be 90% (corresponding to 50 km of fiber) and the initial phase offset to φ_0 to be 0. As expected, when $\eta_{QLP} \leq 1$, $\epsilon \rightarrow 0$ when $B \rightarrow +\infty$. Otherwise, $\epsilon \rightarrow 1 - \frac{1}{\eta_{QLP}}$. There is a compromise between the size of the throttling buffer and the throttling ratio. A larger buffer size implies less release failures, but a longer sojourn time of qubits in memory, resulting in higher decoherence. A lower

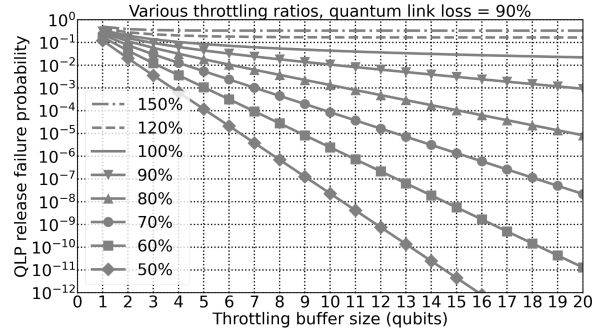


Fig. 4: QLP release failure rate for different throttling ratios.

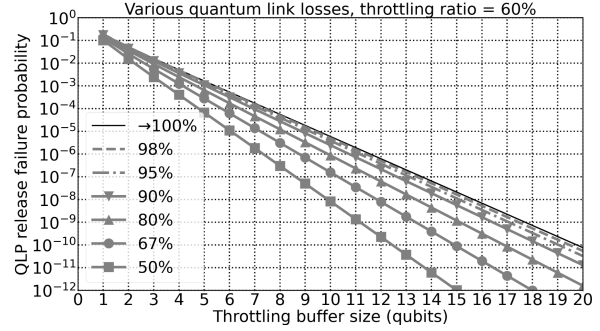


Fig. 5: QLP release failure probability for different losses.

throttling ratio can also reduce release failures, but at the cost of a lower release rate.

Fig. 5 shows the BP release failure probability ϵ for different throttling buffer sizes B and quantum link loss η_{QLL} , taking the throttling ratio η_{QLP} to be 60% and the initial phase offset φ_0 to be 0. The higher the quantum link loss, the higher the QLP release failure ratio. The limit as $\eta_{QLL} \rightarrow 100\%$ is obtained by replacing the binomial distribution with a Poisson distribution of expected value $1/\eta_{QLP}$.

With our Python program, we also find that the BP release failure probability ϵ is independent of the initial phase offset φ_0 between the creation of BPs and the release of qubits. Indeed, when φ_0 varies, the matrices $M^{[\kappa]}$ in Eq. 5 remain identical up to a circular shift of $\Delta\kappa(\varphi_0) \in [0, c-1]$ in the order of multiplication of matrices $M^{(m)}$, which implies that the sequences of matrices $M^{[\kappa]}$ and stationary distributions $\tilde{\pi}_\kappa$ when $\kappa \in [0, c-1]$ are identical for different φ_0 , up to the circular shift of $\Delta\kappa(\varphi_0)$. Thus, Eq. 8 gives the same ϵ .

Conclusions

We propose a throttled QLP that stabilizes the delivery of BPs by each link of an entanglement-based quantum network. Using Markov chains, we model its behavior to quantify its performances. The use of small quantum buffers allows very low release failure probability, at the price of a reasonable reduction in the BP release throughput due to the additional removal of some qubits induced by the throttling process. Our throttled QLP simplifies the processing of entanglement stitching and routing in the upper layers of the quantum network protocol stack, as they receive BPs at fixed rate with very low failure probability ϵ .

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References

- [1] C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing", *Theoretical Computer Science*, vol. 560, pp. 7–11, 2014, Original paper: *Proceedings of the International Conference on Computers, Systems and Signal Processing*, Bangalore, pp. 175–179, Dec. 1984. DOI: 10.1016/j.tcs.2014.05.025
- [2] W. K. Wootters and Z. Wojciech, "A single quantum cannot be cloned", *Nature*, vol. 299, pp. 802–803, Oct. 1982. DOI: 10.1038/299802a0
- [3] C. Wang, A. Rahman, R. Li, M. Aelmans, and K. Chakraborty, "Application scenarios for the quantum internet", RFC Editor, RFC 9583, Jun. 2024, pp. 1–26. [Online]. Available: <http://www.rfc-editor.org/rfc/rfc9583.txt>
- [4] W. Kozłowski, S. Wehner, R. Van Meter, B. Rijsman, A. S. Cacciapuoti, M. Caleffi, and S. Nagayama, "Architectural principles for a quantum internet", RFC Editor, RFC 9340, Mar. 2023, pp. 1–37. [Online]. Available: <http://www.rfc-editor.org/rfc/rfc9340.txt>
- [5] Y. Yu, F. Ma, X.-Y. Luo, B. Jing, P.-F. Sun, R.-Z. Fang, C.-W. Yang, H. Liu, M.-Y. Zheng, X.-P. Xie, W.-J. Zhang, L.-X. You, Z. Wang, T.-Y. Chen, Q. Zhang, X.-H. Bao, J.-W. Pan, "Entanglement of two quantum memories via fibres over dozens of kilometres", *Nature*, vol. 578, pp. 240–245, Feb. 2020. DOI: 10.1038/s41586-020-1976-7
- [6] S. L. N. Hermans, M. Pompili, H. K. C. Beukers, S. Baier, J. Borregaard, and R. Hanson, "Qubit teleportation between non-neighbouring nodes in a quantum network", *Nature*, vol. 605, pp. 663–668, May 2022. DOI: 10.1038/s41586-022-04697-y
- [7] V. Krutyanskiy, M. Galli, V. Krcmarsky, S. Baier, S. D. A. Fioretto, Y. Pu, A. Mazloom, P. Sekatski, M. Canteri, M. Teller, J. Schupp, J. Bate, M. Meraner, N. Sangouard, B. P. Lanyon, T. E. Northup, "Entanglement of trapped qubits separated by 230 meters", *Physical Review Letters*, vol. 130, p. 050803, 5 Feb. 2023. DOI: 10.1103/PhysRevLett.130.050803
- [8] A. J. Stolk, K. L. van der Enden, M.-C. Slater, I. te Raaij-Derckx, P. Botma, J. van Rantwijk, J. J. B. Biemond, R. A. J. Hagen, R. W. Herfst, W. D. Koek, A. J. H. Meskers, R. Vollmer, E. J. van Zwet, M. Markham, A. M. Edmonds, J. F. Geus, F. Elsen, B. Jungbluth, C. Haefner, C. Tresp, J. Stuhler, S. Ritter, and R. Hanson, "Metropolitan-scale heralded entanglement of solid-state qubits", *Science Advances*, vol. 10, no. 44, eadp6442, 2024. DOI: 10.1126/sciadv.adp6442
- [9] J. Illiano, M. Caleffi, A. Manzalini, and A. S. Cacciapuoti, "Quantum internet protocol stack: A comprehensive survey", *Computer Networks*, vol. 213, p. 109092, 2022, ISSN: 1389-1286. DOI: <https://doi.org/10.1016/j.comnet.2022.109092>
- [10] Y. Li, H. Zhang, C. Zhang, T. Huang, and F. R. Yu, "A survey of quantum internet protocols from a layered perspective", *IEEE Communications Surveys & Tutorials*, vol. 26, no. 3, pp. 1606–1634, 2024. DOI: 10.1109/COMST.2024.3361662
- [11] A. Dahlberg, M. Skrzypczyk, T. Coopmans, L. Wubben, F. Rozpundinedek, M. Pompili, A. Stolk, P. Pawelczak, R. Knegjens, J. de Oliveira Filho, R. Hanson, and S. Wehner, "A link layer protocol for quantum networks", in *Proceedings of the ACM Special Interest Group on Data Communication*, ser. SIGCOMM '19, Beijing, China: Association for Computing Machinery, 2019, pp. 159–173. DOI: 10.1145/3341302.3342070
- [12] W. Kozłowski, A. Dahlberg, and S. Wehner, "Designing a quantum network protocol", ser. CoNEXT '20, Barcelona, Spain: Association for Computing Machinery, 2020, pp. 1–16. DOI: 10.1145/3386367.3431293
- [13] Á. G. Iñesta and S. Wehner, "Performance metrics for the continuous distribution of entanglement in multiuser quantum networks", *Physical Review A*, vol. 108, p. 052615, 5 Nov. 2023. DOI: 10.1103/PhysRevA.108.052615
- [14] P. Brémaud, *Markov Chains - Gibbs Fields, Monte Carlo Simulation and Queues*, 2nd edition. Springer Nature Switzerland AG, 2020. DOI: 10.1007/978-3-030-45982-6